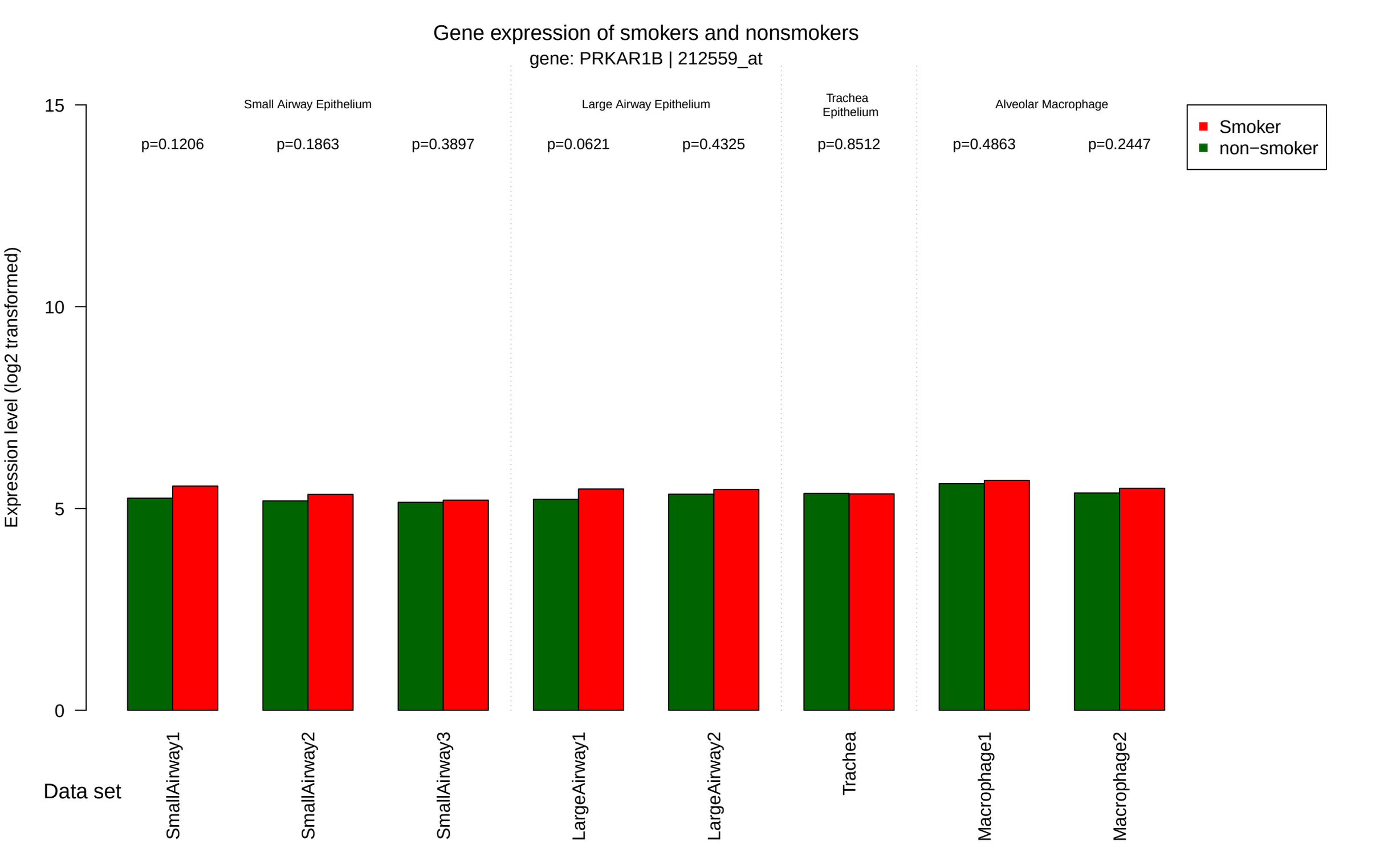
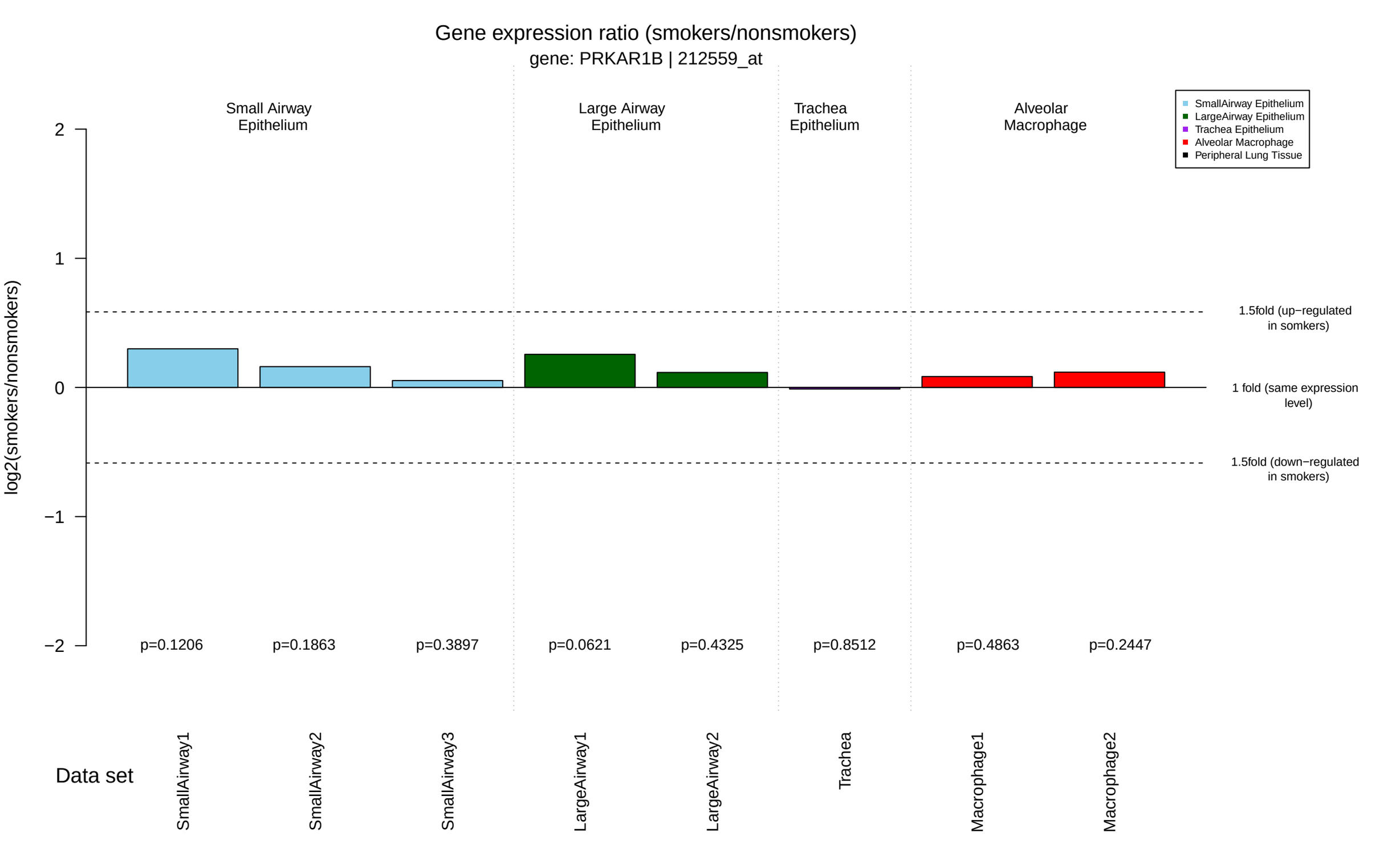
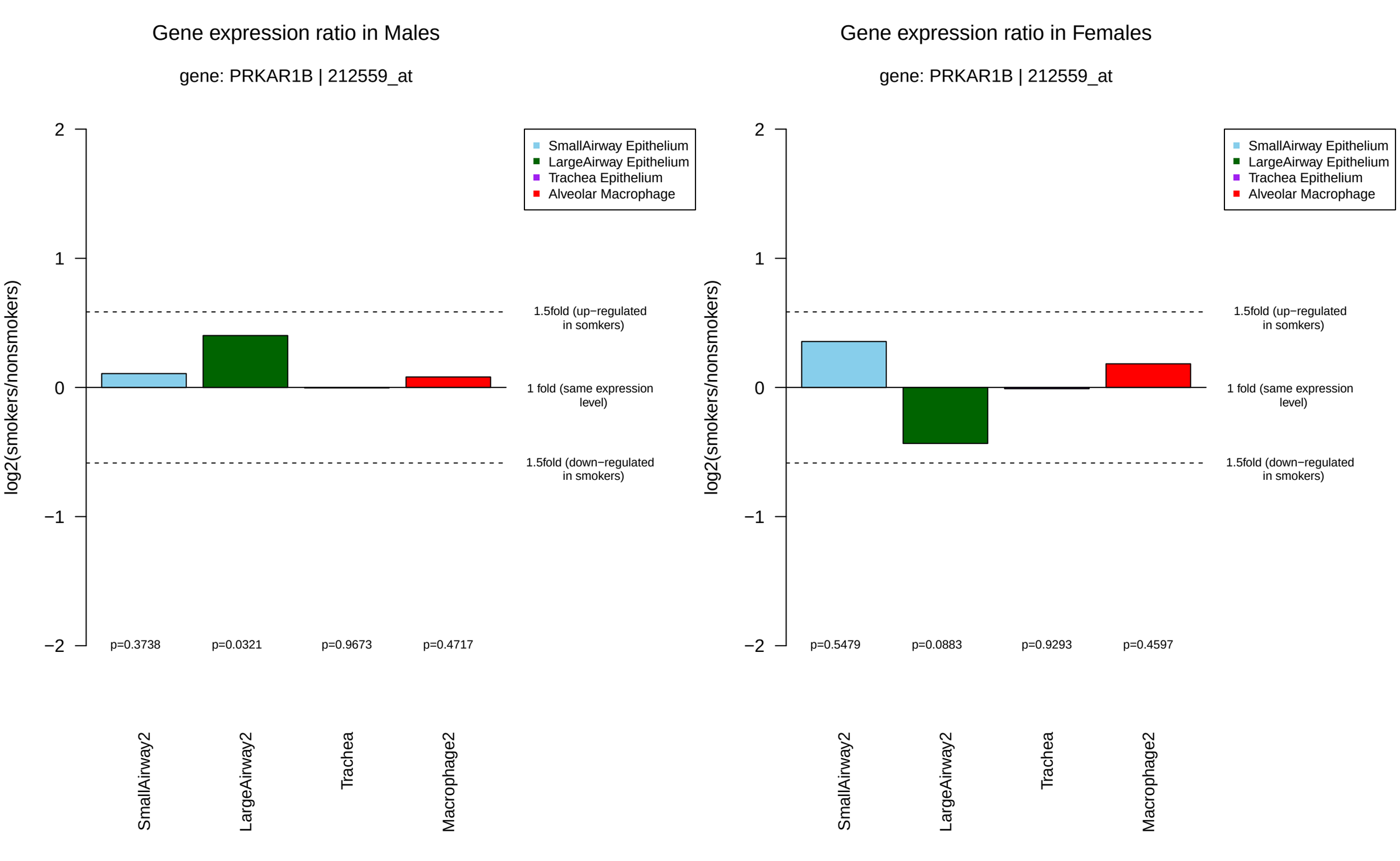
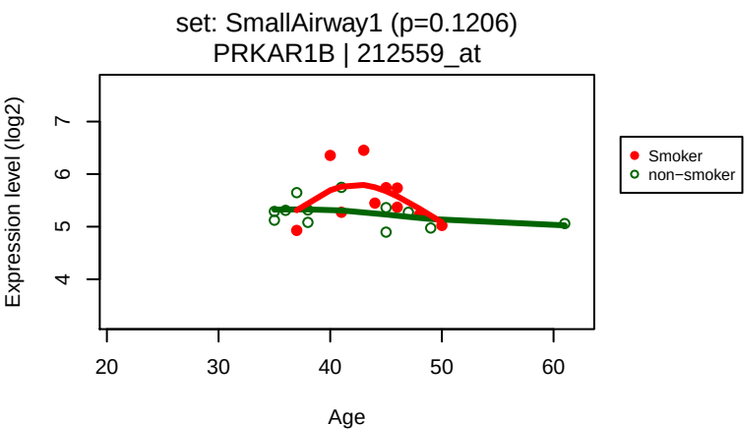
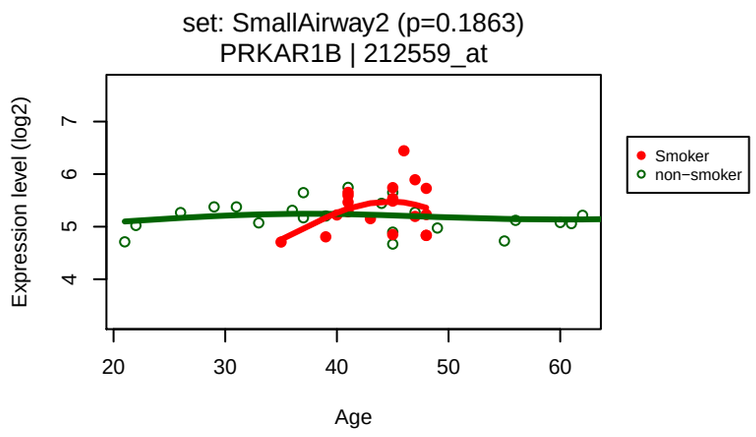
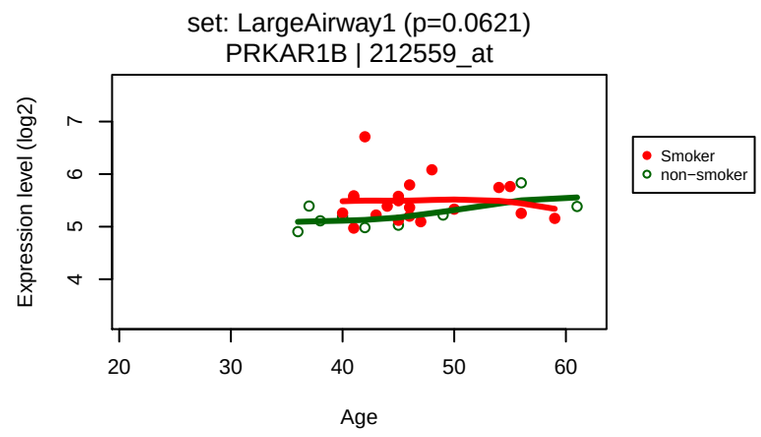

Gene expression of smokers and nonsmokers
gene: PRKAR1B | 212559_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1206
p=0.1863
p=0.3897
p=0.0621
p=0.4325
p=0.8512
p=0.4863
p=0.2447
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

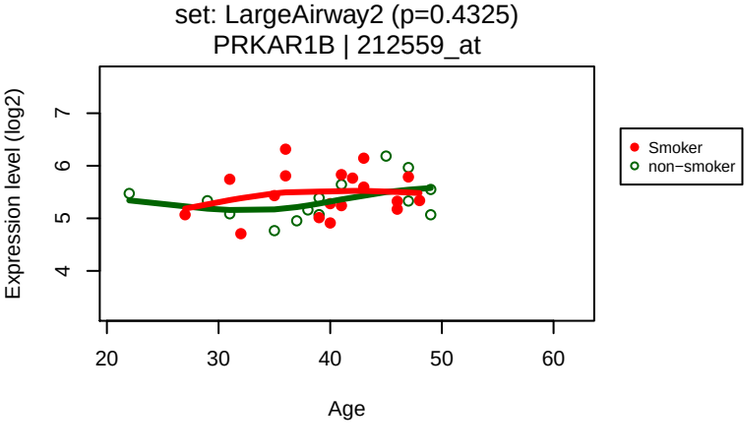

Gene expression ratio (smokers/nonsmokers)
gene: PRKAR1B | 212559_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1206
p=0.1863
p=0.3897
p=0.0621
p=0.4325
p=0.8512
p=0.4863
p=0.2447
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

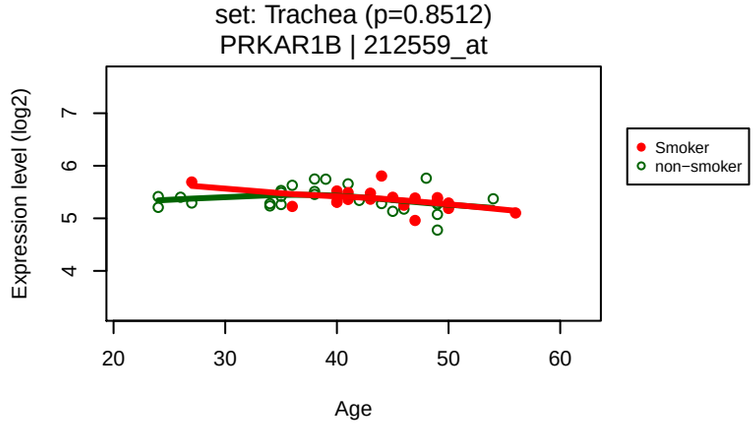

Gene expression ratio in Males
gene: PRKAR1B | 212559_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3738
p=0.0321
p=0.9673
p=0.4717
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAR1B | 212559_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5479
p=0.0883
p=0.9293
p=0.4597
SmallAirway2
LargeAirway2
Trachea
Macrophage2

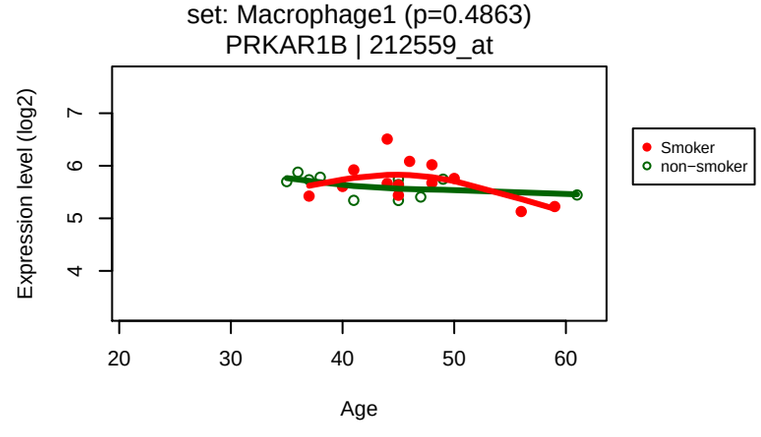

set: SmallAirway1 (p=0.1206)
PRKAR1B | 212559_at
Expression level (log2)
Age
Smoker
non−smoker

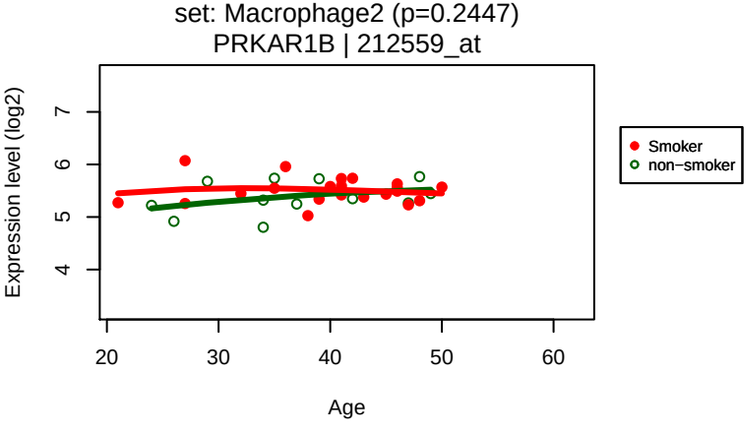

set: SmallAirway2 (p=0.1863)
PRKAR1B | 212559_at
Expression level (log2)
Age
Smoker
non−smoker

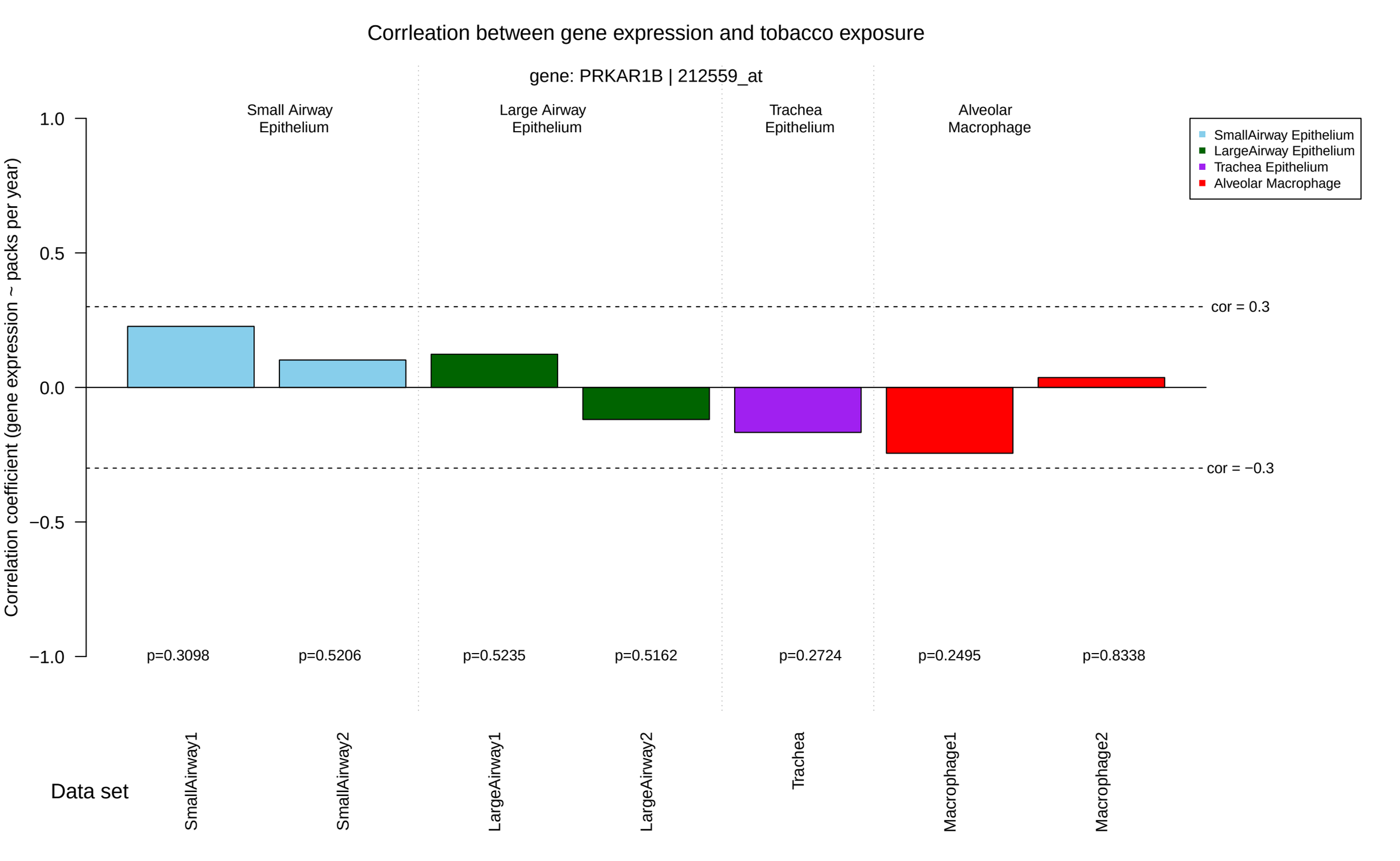

set: LargeAirway1 (p=0.0621)
PRKAR1B | 212559_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4325)
PRKAR1B | 212559_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8512)
PRKAR1B | 212559_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4863)
PRKAR1B | 212559_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2447)
PRKAR1B | 212559_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAR1B | 212559_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3098
p=0.5206
p=0.5235
p=0.5162
p=0.2724
p=0.2495
p=0.8338
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2